

E-Commerce Database Management System

Project Report

A relational database modeling an online store's catalog, customers, orders, payments, shipping, and reviews — built in PostgreSQL.

1. Project Overview

This project implements the backend database for an e-commerce platform. It covers the full order lifecycle — a customer browses a categorized product catalog, places an order made up of one or more line items, pays for it, has it shipped, and can later leave a review. The schema is normalized to third normal form, with foreign keys and CHECK constraints enforcing data integrity throughout.

Scope

Tables	9
Customers	100
Product categories	10
Products	200
Orders	300
Order line items	700
Payments / Shipping records	300 / 300
Reviews	500
Business queries	28
Reporting views	7

2. Entity-Relationship Diagram

The diagram below shows all nine tables, their key columns, and the relationships between them.

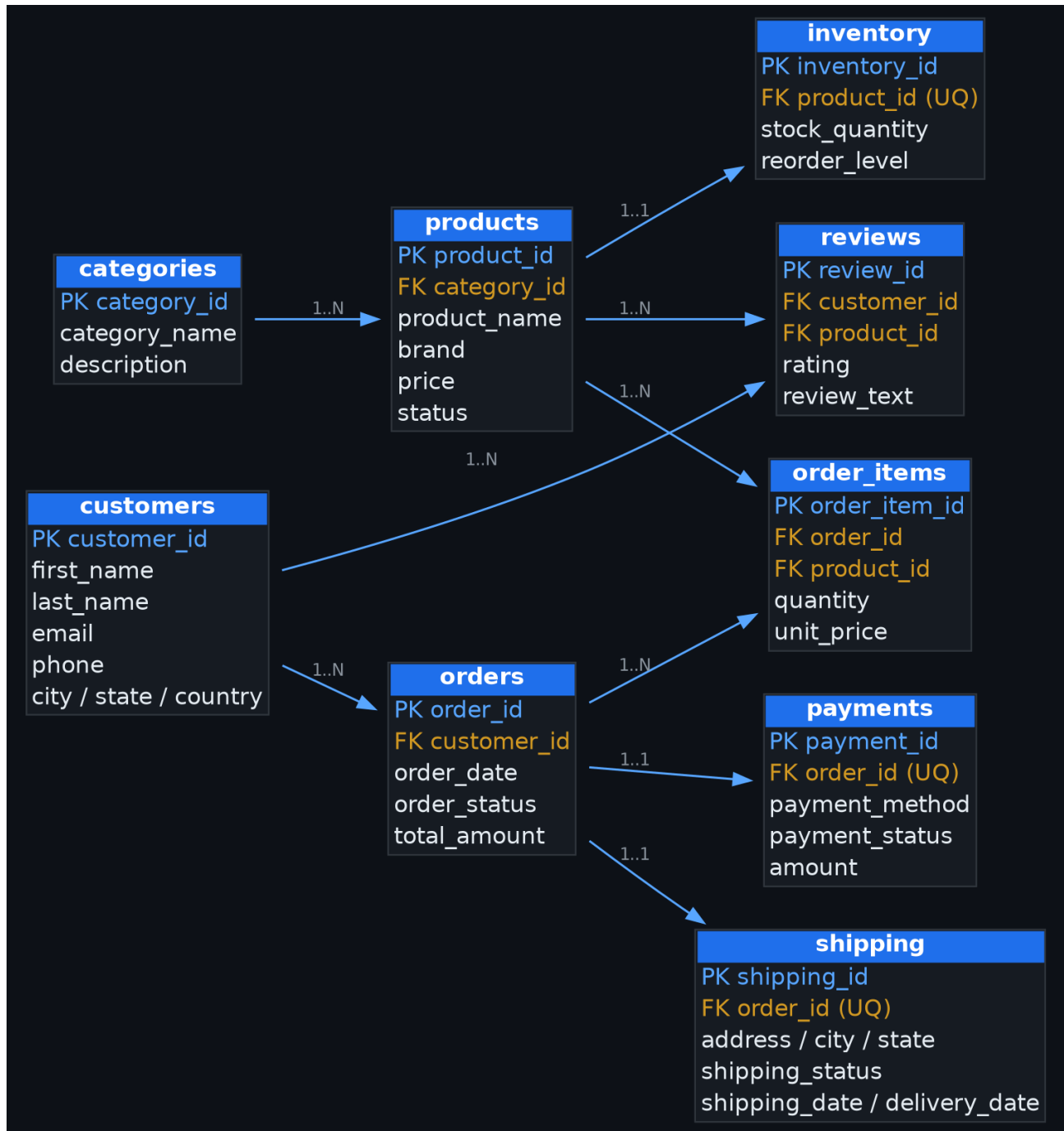


Figure 1: Entity-relationship diagram of ecommerce_db

Schema Overview

Grouped at a higher level, the schema separates cleanly into three concerns: the product catalog, the customer, and the transaction (order, payment, shipping).

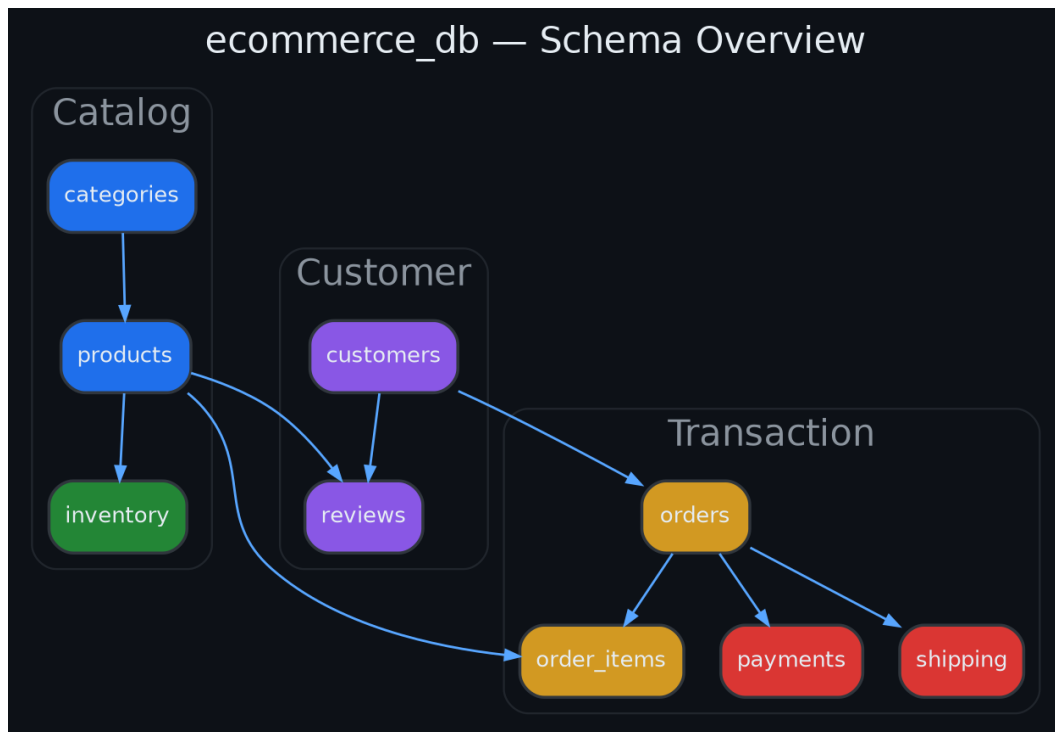


Figure 2: High-level schema grouping

3. Key Design Decisions

Inventory is separate from products

Stock levels change far more often than catalog data, so keeping them in their own table avoids unnecessary writes to the product record.

Order line items store their own unit_price

This freezes the price at the moment of purchase, so a later price change on a product does not retroactively alter historical invoices.

Payments and shipping are split from orders

Each is a 1:1 relationship today, but keeping them separate keeps each table focused on one concern and leaves room to extend either independently.

All monetary columns use NUMERIC(10,2)

Avoids the floating-point rounding errors that come with FLOAT/REAL types for currency.

Status columns are constrained with CHECK

order_status, payment_status, shipping_status, and product status are all restricted to a fixed set of values at the database level.

4. Sample Query Outputs

ecommerce_db – Tables

name
categories
customers
inventory
order_items
orders
payments
products
reviews
shipping

Figure 3: All tables in ecommerce_db

Order → Customer → Product Join

order_id	customer_name	product_name	quantity	unit_price	line_total	order_status
1	Rohan Kumar	Product 110	5	56690.93	283454.65	Pending
1	Rohan Kumar	Product 103	5	9435.14	47175.7	Pending
1	Rohan Kumar	Product 13	1	4028.73	4028.73	Pending
2	Kabir Jha	Product 137	1	37652.12	37652.12	Delivered
3	Kabir Singh	Product 117	1	83087.25	83087.25	Processing
4	Neha Verma	Product 160	4	78179.13	312716.52	Processing
5	Arjun Verma	Product 29	4	28200.42	112801.68	Shipped
5	Arjun Verma	Product 66	4	72662.24	290648.96	Shipped
5	Arjun Verma	Product 166	1	48963.69	48963.69	Shipped
5	Arjun Verma	Product 130	2	1570.24	3140.48	Shipped

Figure 4: Order → customer → product join

View: product_catalog

product_id	product_name	brand	category_name	price	status	stock_quantity	reorder_level	stock_status
1	Product 1	Apple	Fashion	13350.88	Available	196	10	In Stock
2	Product 2	LG	Furniture	83072.17	Available	75	10	In Stock
3	Product 3	Boat	Furniture	13257.24	Available	83	10	In Stock
4	Product 4	HP	Toys	73931.34	Available	292	10	In Stock
5	Product 5	Puma	Automotive	70974.41	Available	29	10	In Stock
6	Product 6	Apple	Books	5849.43	Available	79	10	In Stock
7	Product 7	Dell	Beauty	16731.01	Available	34	10	In Stock
8	Product 8	Nike	Home & Kitchen	27804.81	Available	207	10	In Stock
9	Product 9	Samsung	Furniture	3729.03	Available	73	10	In Stock
10	Product 10	HP	Books	11243.95	Available	141	10	In Stock

Figure 5: product_catalog view

Monthly Revenue Report

order_month	total_orders	total_revenue	avg_order_value
2025-01	19	516552.45	27186.97
2025-02	18	400576.46	22254.25
2025-03	17	472499.01	27794.06
2025-04	20	642392.91	32119.65
2025-05	18	441513.31	24528.52
2025-06	13	400731.55	30825.5
2025-07	24	567918.2	23663.26
2025-08	15	413518.04	27567.87
2025-09	20	527512.31	26375.62
2025-10	21	491628.61	23410.89
2025-11	17	471732.34	27748.96
2025-12	23	573244.3	24923.67

Figure 6: Monthly revenue report

Top 10 Customers by Spend

customer_id	first_name	last_name	total_spent
99	Ananya	Verma	3350060.86
74	Aditi	Yadav	3095353.45
4	Vivaan	Singh	2868643.39
55	Kabir	Joshi	2776052.77
86	Aarav	Jha	2400462.68
76	Karan	Gupta	2276968.37
9	Rahul	Patel	2259883.52
85	Rohan	Singh	2187949.86
39	Aditya	Sharma	2142906.24
10	Meera	Patel	2110506.22

Figure 7: Top customers by lifetime spend

Low Stock / Inventory Alert

product_name	stock_quantity	reorder_level
Product 57	0	10
Product 121	0	10
Product 163	0	10
Product 32	1	10
Product 94	1	10
Product 143	2	10
Product 97	5	10
Product 43	8	10
Product 118	8	10
Product 200	9	10

Figure 8: Low-stock inventory alert

5. Conclusion

The resulting schema supports realistic e-commerce reporting out of the box: revenue trends, top customers, inventory alerts, delivery performance, and product ratings, all backed by enforced referential integrity. The seven views in 05_views.sql package these as reusable building blocks for a dashboard or BI layer on top of the raw tables.